

Luca Walz

Backend and Cloud Engineer

GitHub github.com/lucawalz · LinkedIn linkedin.com/in/luca-walz · lucawalz.dev

Backend and cloud engineer who likes following a system the whole way down, from the API surface to the infrastructure underneath. Most of my work sits around Kubernetes, NixOS and GitOps, with a fair bit of Go, plus LLM agents and full-stack IoT on the side.

EXPERIENCE

IBM — Böblingen (Hybrid)

Working Student (Part-time)

Software Development Intern (Full-time)

10 months

Sep 2025 - Dec 2025

Mar 2025 - Aug 2025

Built API test suites and backend testing workflows for IBM Knowledge Catalog, validating enterprise-scale microservices alongside the dev and QA teams. (Java, Jenkins)

EDUCATION

Hochschule der Medien, Stuttgart — BSc Media Informatics

Bachelor thesis on autonomous cluster remediation with LLM agents.

2023 - Present

Freie Waldorfschule, Böblingen — Abitur

2020 - 2022

Burnaby Mountain Secondary School, Vancouver — Exchange year, Canada

2019 - 2020

SELECTED PROJECTS

vigil Python, Go

Multi-agent system that diagnoses and repairs Kubernetes faults on its own through typed MCP tools, reversible by construction.

bedrock Nix

Bare-metal Kubernetes homelab that lives entirely in Git; three NixOS nodes reconciled by Flux.

horizon Go

Controller that watches K3s for resource pressure and bursts onto Hetzner Cloud over a WireGuard overlay, then tears the node down after the spike.

sentio-systems Java, TypeScript, Python

Wildlife and weather monitoring; Raspberry Pi field devices feed a Spring Boot backend with AI classification and a React dashboard.

SKILLS

Languages

Go, Python, Java, TypeScript

Runtimes & frameworks

Spring, React, Nix

Infrastructure & data

Linux, Kubernetes, Docker, Terraform, Postgres, Redis, Prometheus, Grafana, Raspberry Pi

VOLUNTEERING

Student Mentor — Hochschule der Medien, Stuttgart

Supporting first-semester students in finding their footing on campus and in their studies.

Mar 2024 - Aug 2024

Scout

Long-term member, organising outdoor excursions with larger groups.

Since 2008

Downhill Trainer

Since 2014

LANGUAGES

German (Native) · English (C2) · Russian (A1)